Bridging Linear Algebra and Quantum Mechanics

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Abstract

Quantum mechanics utilizes the concepts of linear algebra to represent the fundamental laws of nature on the atomic and subatomic scales. The space of states of any quantum system can be represented as a complex Hilbert space equipped with a Hermitian inner product. With the mathematical tools provided in linear algebra and the intuitions provided by physics, we will rebuild the framework of quantum mechanics in this paper, providing a comprehensive dictionary of these terms as we progress. Ultimately, we want to show the robustness of this framework by proving the Time-Dependent Schrödinger Equation.

1 Why Linear Algebra?

Before we proceed with the math, we must first justify our choice of using Linear Algebra to represent the quantum world. To do so, we must compare our understandings of classical and quantum physics:

1.1 Classical Physics versus Quantum Mechanics

Let us compare quantities between the two disciplines. In Classical Physics, quantities (like mass, momentum, and position) are **single-valued** and **continuous**, taking on only a single value at any time and varying smoothly over time. Thus, we can denote an object or system with **continuous functions**, with common parameters being position and time.

However, quantities in quantum mechanics behave fundamentally differently. There are many examples, but take, for example, an excited hydrogen atom. We observe that as the electron in the atom orbits and falls toward the proton, it will release energy in the form of light. Under the classical interpretation, the light released can be modeled as a continuous function of energy over time. However, when preformed experimentally, we only read a few "blips" from our detectors, measuring a discrete number of quantities. Repeating this experiment, we may obtain a different set of observed values, but upon repeating further, it appears we only get a **discrete** set of possible energy values where the specific energy values we measure are **non-uniformly random**.

1.2 Proposed Framework

Thus, we find that, unlike classical mechanics, quantum physical quantities can be discrete and random but probabilistic. To model this, we need means by which to combine our possible outcomes into one aggregate object that describes our particle before a measurement that encodes the probability and discreteness of our values. Hence, we can use the objects of Linear Algebra to model our desired properties. Using the apparent analogs, superposition states translate to linear combinations, where **quantum states** are vectors (with the added stipulation that only direction contributes to the state), and physical quantities can be expressed as linear functionals (named observables) acting on our state vectors. As we continue, we will further flesh out these definitions and expand on this proposed framework.

2 Dirac Notation, Wave Functions, and the Hilbert Space

2.1 Dirac Notation

Dirac Notation, also known as **Bra-Ket notation**, was a notation system developed by Paul Dirac and published in 1939 as a way to simplify the writing of linear algebra concepts in quantum mechanics.

2.1.1 Kets

Kets represent our vectors in quantum mechanics. Thus, they adopt all the same properties we would expect a vector to have. These vectors represent quantum states which hold the data encoding all the physical properties of our particular particle. We express the ket of some vector ψ as $|\psi\rangle$.

With a definition partially set, we can use our understanding of definite states to find a superposition ket. A **definite states** is the quantum state attributable to a certain measurement. For example, let's consider $|E_i\rangle$ to be the quantum state associated with the energy reading E_i, having an associated coefficient of c_i which is related to our probability. We can express the energy superposition of n number of energy quantum states to be:

$$|\psi\rangle = \sum_{i=0}^{n} c_i \, |E_i\rangle$$

The reason why we can't explicitly express c_i as the probabilities of our quantum states is because of **Born's Rule**, which states that the real value probability of measuring any definite state is $p_i = |c_i|^2 / |\,|\psi\rangle\,|^2$, where $|c_i|$ is the complex magnitude of c_i and $|\,|\psi\rangle\,|$ is the length of our given quantum state (read inner products for definition of length). While the proof isn't difficult, for the sake of brevity, read **Gleason's Theorem** which proves such relationship. Thus, we will add a stipulation that any state ket will be normalized (and consequently, the direction of the ket defines the state, not the length). This is investigated further in the section titled Inner Products. Note, that while c_i is directly related to our probability, it is a complex probability value. It is an important distinction to aknowledge.

Suppose, now that there exist an infinitely many number of state kets in our linear combination (e.g. a decomposition of a wave into an infinite number overtones). This poses an issue as under any arbitrary complex vector space, we may not have completeness. Thus, we must consider the vector space upon which our kets belong.

2.1.2 Hilbert Spaces

As previously mentioned, our vector space must support an infinite linear combination. Thus, we must consider a vector space with an infinite number of basis vectors and thus infinite dimension. (While it isn't necessary that an infinite linear combination implies an infinite dimensionality, when we introduce eigenstates, inner products, and orthogonality we will see why it is implied). But, without the necessary stipulation of **Cauchy completeness**, that every convergent sum in our vector space must converge to an element inside the vector space, we may end up with an element outside it (e.g. take an infinite combination of elements in $\mathbb{R}$[X] with an infinitely supported linear combination, we can obtain a function like $f(x) = e^x$ with Taylor Expansions).

Further, we want to introduce inner products into our vector space because it allows us to practically define length and other vital properties into our framework. The vector space we described here is by definition known as the **Hilbert Space**, $\mathcal{H}$. Note, while we are contained within the Hilbert Space, in non-infinite cases, we work in subspaces of $\mathcal{H}$.

2.1.3 Inner Product

The **Inner Product** is a bilinear map that inputs two vectors and outputs a scalar value ($< \mid > : V \times V \to F$), acting as a much more generalized form of the dot product. Notated as $\langle\phi|\psi\rangle$, the inner product is useful for many reasons. It allows us to extend the notion of angles of vectors into our vector

space, allowing us to define orthogonality (where $|\phi\rangle \perp |\psi\rangle$ if and only if $\langle\phi|\psi\rangle = 0$). Furthermore, we can define the length of a ket $||\psi\rangle|$ as the inner product of the vector with itself $\langle\psi|\psi\rangle$. Since $\mathcal{H}$ is over the field $\mathbb{C}$ and we want a positive $\mathbb{R}$ value to define our length, we need to add a few restrictions on our inner product.

I) **Inner Products Should be Linear in the Right-Hand Position**: For $|\psi\rangle, |\phi\rangle, |\zeta\rangle$, and scalar a, $\langle\psi|\phi + \zeta\rangle = \langle\psi|\phi\rangle + \langle\psi|\zeta\rangle$ and $\langle\phi|a\psi\rangle = a\langle\phi|\psi\rangle$

II) $\langle\psi|\phi\rangle = \langle\phi|\psi\rangle^*$: The reason why we add this antisymmetric condition is because if we have some inner product $\langle\phi|\phi\rangle$=1, we have a real length. However, now suppose we have, $\langle i\phi|i\phi\rangle$. If we have a strictly bilinear inner product, this would mean we would have a defined length of $||i\phi\rangle| = i$. Thus, we must add the antisymmetric condition to ensure that we preserve real positive length ($\langle i\phi|i\phi\rangle = i\langle\phi|i\phi\rangle^* = ii^*\langle\phi|\phi\rangle^* = ii^* = 1$

III) **Only the Zero Vector has 0 Length**: $|\psi\rangle = 0 \iff \langle\psi|\psi\rangle = 0$.

This is by definition the properties of the **Hermitian Inner Product**, meaning that the Hilbert space on which we are working is also a **Hermitian Inner Product Space.**

While we used it above, we can now formally define the length of a vector, $||\psi\rangle| = \sqrt{\langle\psi|\psi\rangle}$, and orthogonality of two vectors as $\langle\psi|\phi\rangle = 0$. Now suppose we have an orthonormal basis for our vector space, $\beta = \{|E_i\rangle\}$. This means that $\langle E_i|E_i\rangle = 1$ and $\langle E_i|E_j\rangle = 0\ \forall i \neq j$. We can summarize this with the **Kronecker Delta**:

$$\langle E_i|E_j\rangle = \delta_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$$

So, if we have an orthonormal basis and expand some arbitrary quantum state in this, $|\psi\rangle = \sum_{i=1}^{n} c_i |E_i\rangle$, we want to identify the values our c's. We discussed earlier that our c's under this orthonormal basis are our probabilities, but we can use the inner product to explicitly solve for them. For the probability of some desired state, j:

$$\langle E_j|\psi\rangle = \langle E_j|\sum_{i=1}^{n} c_i |E_i\rangle = \sum_{i=1}^{n} c_i \langle E_j|E_i\rangle = c_j\delta_{jj} = c_j$$

(While we split the bra-ket into a bra and a ket, this is sound by the Riesz Representation Theorem discussed in the section title Bras.)

Moreover, we can find an explicit form of the inner product in our Hilbert Space. Suppose we have two kets $|\psi\rangle, |\phi\rangle \in \mathcal{H}$, and a basis $\beta = \{|E_i\rangle\}$:

$$\langle\psi|\phi\rangle = (\sum_{i=1}^{n} \langle a_i E_i|)(\sum_{j=1}^{n} |b_j E_j\rangle = (\sum_{i=1}^{n} a^* \langle E_i|)(\sum_{j=1}^{n} b_j |E_j\rangle =$$

$$\sum_{i=1}^{n}\sum_{j=1}^{n} a_i^* b_j \langle E_i|E_j\rangle = \sum_{i=1}^{n}\sum_{j=1}^{n} a_i^* b_j \delta_{ij} = \sum_{i=1}^{n} a_i^* b_i$$

This is the explicit form of the inner product over the complex Hilbert Space is our "dot product." In fact, if we restrict our kets to only be over the real Hilbert Space, our formula for the inner product becomes the Euclidean dot product (since then $a_i^* = a_i$). This is a powerful tool, and we will be returning to it often.

2.1.4 Bras

Bras notated as $\langle\psi|$ are elements in the dual space of our Hilbert Space, $\mathcal{H}^*$. By definition, it is a linear functional, taking a vector in our Hilbert Space and mapping it to some scalar in $\mathbb{C}$. In Quantum Mechanics, it is often used to extract of a single, complex number representing the "probability amplitude" of a given quantum state.

By observation, the action of a bra acting on a ket holds many similarities to an inner product.

In fact, by the **Riesz Representation Theorem**, they are interconnected. It states that the action of some linear functional on some ket $|\psi\rangle$ is the equivalent to taking the inner product with some unique ket and $|\psi\rangle$:

$$f : \mathcal{H} \to \mathbb{C} \implies \exists! \, |\phi\rangle \text{ s.t } \langle\phi|\psi\rangle = f(|\psi\rangle)$$

In Dirac Notation, this functional f is notated as the bra $\langle\phi|$, so we have that $\langle\phi|\psi\rangle = \langle\phi| \, |\psi\rangle$. Despite being two seperate concepts, linear functionals and inner products are conveniently related in Bra-Ket notation. This leads us to an important resolution in our theory (assuming we have some ket $|\psi\rangle$ and basis $\beta = \{A_i\}$:

$$|\psi\rangle = \sum_{i=1}^{n} c_i \, |A_i\rangle = \sum_{i=1}^{n} \langle A_i|\psi\rangle \, |A_i\rangle = \sum_{i=1}^{n} \langle A_i| \, |\psi\rangle \, |A_i\rangle = \sum_{i=1}^{n} |A_i\rangle \, \langle A_i| \, |\psi\rangle = (\sum_{i=1}^{n} |A_i\rangle \, \langle A_i|) \, |\psi\rangle$$

$$\implies \hat{I} = \sum_{i=1}^{n} |A_i\rangle \, \langle A_i|$$

This is known as the resolution of the identity. It makes sense intuitively as if we apply some ket $|\phi\rangle$ to either side, we get $|\phi\rangle$ in our basis β.

2.2 Wavefunctions and Continuous Quantities

Thus far, we have only considered the cases when quantities have a discrete set of possible values. However, not all quantities are discrete such as position states. We resolve this by combining the concepts of linear superpositions and calculus. Hence, for a continuous set of state kets $\{|x\rangle\}$, we can use an integral to express a linear combination:

$$\phi = \int_{-\infty}^{\infty} \psi(x) \, |x\rangle \, dx \approx \sum_{i=1}^{n} c_i \, |E_i\rangle$$

Analyzing it piecemeal, $\psi(x)$ is our coefficients as a continuous function of position; this is known as the wave function. $|x\rangle$ represents our state kets. And, dx represents the parameter upon which we are integrating over.

2.2.1 Dirac Deltas Functions

Suppose we have a continuous position orthonormal basis. Like for the discrete case, we want to find $\psi(x)$ for a given position quantum state corresponding to position a. We can begin by taking the inner product:

$$\langle a|\phi\rangle = \langle a| \int_{-\infty}^{\infty} \psi(x) \, |x\rangle \, dx = \int_{-\infty}^{\infty} \psi(x) \, \langle a|x\rangle$$

Given our orthonormal basis, we want to represent our $\langle a|x\rangle$. For this, we can use the Dirac Delta Function, $\delta(x)$. While there are arguments over the precise nature of this function, it is still widely used in mathematics and physics. Most intuitively think of the function as a "big spike" where $f(x) = 0$, except for at $x = 0$, where $f(x) = \infty$. However, this is not mathematically sound. For the purposes of this paper, we will circumvent these debates by saying the definition of this function is $f(c) = \int_{-\infty}^{\infty} f(x)\delta(x - c)dx$. Consequently, the equivalent to the Kronecker delta for the continuous case is $\delta(x_i - x_j) = \langle x_i|x_j\rangle$ Thus, if we apply this to our previous equation, we get

$$\langle a|\phi\rangle = \int_{-\infty}^{\infty} \psi(x) \, \langle a|x\rangle = \int_{-\infty}^{\infty} \psi(x)\delta(x - a) = \psi(a)$$

Thus, we have a method by which we can obtain our wavefunction. We can further extend these properties to determine the inner product of two continuous state kets $|\psi\rangle$ and $|\phi\rangle$:

$$\langle\psi|\phi\rangle = (\int_{-\infty}^{\infty} \psi(x) \, \langle x| \, dx)(\int_{-\infty}^{\infty} \phi(y) \, |y\rangle \, dy) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \psi^*(x)\phi(y)\delta(x - y)dydx = \int_{-\infty}^{\infty} \psi^*(x)\phi(x)dx$$

3 Observables and Linear Operators

We posited that physical quantities in our framework were obtained by applying a linear operator on our Hilbert Space. These quantities include position, momentum, energy, and angular momentum as a few examples. Because Quantum Mechanics rarely ever uses a standard basis, we will stray away from any matrix representation of our linear operator. In this section, we will be discussing why linear operators are the desired tool to express **Observables**, a physical property or quantity able to be measured.

3.1 Eigenstates

As we have discussed, definite states are states in which we have a precisely defined physical value for an observable. So, we know both the direction of the state and its magnitude. Thus, when we act an observable on any of these defined states, they should maintain the direction of the state ket and have a magnitude corresponding the quantity we are trying to measure. Mathematically speaking, this appears similar to the definition of an eigenvector and eigenvalue respectively: $\hat{O}\left|\psi\right\rangle = \lambda\left|\psi\right\rangle$, where $\hat{O}$ is our observable. These definite states are known as **Eigenstates** of the observable. Furthermore, we desire that when we apply an observable on a state ket, it acts the observable on each of the eigenstates in the ket's linear combination, as such:

$$\hat{O}\left|\psi\right\rangle = \sum_{i=1}^{n} c_i \hat{O}\left|E_i\right\rangle = \sum_{i=1}^{n} c_i \lambda_i \left|E_i\right\rangle$$

In summary, we have that our observable maps vectors to other vectors in our Hilbert Space, and it preserves the properties of addition and scalar multiplication on our vector space. Thus, our observables satisfy the definition of a linear transformation. Moreover, our observables and eigenstates hold specific properties:

I) **Observables Must Have Real Eigenvalues**: Physical quantities don't hold complex values (e.g. Energy $\neq 5 + 2i$ eV).

II) **An Observable's Eigenstates Must Span Our Vector Space**: Every particle has a value for an observable since there exist no particles with value = "null." Suppose our eigenstates did not span the entirety of the vector space. That implies there exists a quantum state outside the span of our eigenstates and thus cannot be expressed as a linear combination of our eigenstates. Consequently, because our observables act only on eigenstates, our quantum state has no possible value outcomes, posing a contradiction.

III) **Eigenstates are Mutually Orthogonal**: Suppose we have $\left|L_1\right\rangle \not\perp \left|L_2\right\rangle$; we can decompose $\left|L_1\right\rangle = \alpha\left|L_2\right\rangle + \beta\left|L_2^{\perp}\right\rangle$. This means our definite state is linearly dependent on another definite state, which poses a contradiction, as a definite state cannot be in a superposition of definite states.

IV) As a consequence of II), III), and the fact that our eigenstates are normalized, we have that our set of definite states forms an orthonormal basis for our vector space.

3.2 Hermitian Observables

To begin this section, we will introduce the concept of a **Hermitian Adjoint**. Suppose we have some kets $\left|\psi\right\rangle, \left|\phi\right\rangle$ and an operator $\hat{M}$. The Hermitian Adjoint of $\hat{M}$, notated $\hat{M}^{\dagger}$, is an operator such that $\langle\psi|\hat{M}\phi\rangle = \langle\hat{M}^{\dagger}\psi|\phi\rangle$. We can consider $\dagger : End(V) \to End(V)$. From this, we can define a **Hermitian Operator**: it an operator that satisfies $\hat{H} = \hat{H}^{\dagger} \iff \langle\hat{H}(\psi)|\phi\rangle = \langle\psi|\hat{H}(\phi)\rangle$. With this definition, we can establish a few important properties for the Hermitian Adjoint:

-) $(\hat{H}^{\dagger})^{\dagger} = \hat{H}$
-) $(\hat{A}\hat{B})^{\dagger} = \hat{B}^{\dagger}\hat{A}^{\dagger}$
-) $\left|\psi\right\rangle^{\dagger} = \langle\psi|$ and $\langle\psi|^{\dagger} = \left|\psi\right\rangle$
-) $(\hat{A} + \hat{B})^{\dagger} = \hat{A}^{\dagger} + \hat{B}^{\dagger}$
-) $c^{\dagger} = c^{*}$
-) $\langle\phi|\psi\rangle^{\dagger} = \left|\psi\right\rangle^{\dagger}\langle\phi|^{\dagger}$

Proof of these properties is left as a simple exercise for the reader.

Let us now consider what role the Hermitian Adjoint play into our theoretical framework. We can express an observable, $\hat{A}$, as the following (also suppose we have some arbitrary ket $|\psi\rangle$ over our basis of eigenstates $\beta = \{E_i\}$ and set of corresponding eigenvalues $\beta_\lambda = \{\lambda_i\}$):

$$\hat{A}|\psi\rangle = \hat{A}(\sum_{i=1}^{n} c_i |E_i\rangle = \sum_{i=1}^{n} c_i \hat{A}|E_i\rangle = \sum_{i=1}^{n} \langle E_i|\psi\rangle \lambda_i |E_i\rangle = \sum_{i=1}^{n} \lambda_i |E_i\rangle \langle E_i| |\psi\rangle = (\sum_{i=1}^{n} \lambda_i |E_i\rangle \langle E_i|) |\psi\rangle$$

$$\implies \hat{A} = \sum_{i=1}^{n} \lambda_i |E_i\rangle \langle E_i| \approx \hat{x} = \int_{-\infty}^{\infty} \lambda(x) |x\rangle \langle x| \, dx$$

Above are the expansions for an observable in terms of its eigenvalues and eigenstates for both the discrete and continuous cases. Now, if we apply the Hermitian adjoint to our expansion, we have:

$$\hat{A}^\dagger = (\sum_{i=1}^{n} \lambda_i |E_i\rangle \langle E_i|)^\dagger = \sum_{i=1}^{n} \lambda_i^\dagger \langle E_i|^\dagger |E_i\rangle^\dagger = \sum_{i=1}^{n} \lambda_i^* |E_i\rangle \langle E_i|$$

$$\text{Since } \lambda_i \in \mathbb{R}+$$

$$= \sum_{i=1}^{n} \lambda_i |E_i\rangle \langle E_i| = \hat{A}^\dagger = \hat{A}$$

This is a significant development as all observables we are working with are indeed Hermitian Operators. Thus, we can use their properties as well.

3.3 Observable Commutators and the Uncertainty Principle

We say that if $\hat{A}\hat{B} = \hat{B}\hat{A}$, then the operators **"commute"**. If two operators commute, then that allows us much greater mathematical freedom as to how we can solve, expand, and prove properties and equations. This motivates the definition of a **Commutator**:

$$\text{Given two operators } \hat{A}, \hat{B}, \text{ their commutator is}$$
$$[\hat{A}, \hat{B}] := \hat{A}\hat{B} - \hat{B}\hat{A} \implies (\hat{A}, \hat{B} \text{ commute} \iff [\hat{A}, \hat{B}] = 0)$$

We want to observe the effect of commutators in our framework, so let us suppose that we have $\hat{A}, \hat{B}$ that are physical observables, meaning they both have an orthonormal basis, and they commute. We want to see how an eigenstate of $\hat{A}$ corresponds to an eigenstate of $\hat{B}$.

Let $\beta = \{E_i\}$ be the eigenstate basis for $\hat{A}$ and $\beta_\lambda = \{\lambda_i\}$ be the corresponding set of eigenvalues. For some $|E_i\rangle \in \beta, \hat{A}|E_i\rangle = \lambda |E_i\rangle$. Now, consider $\hat{B}\hat{A}|E_i\rangle = \hat{B}(\lambda_i |E_i\rangle) = \lambda_i \hat{B}|E_i\rangle = \hat{A}\hat{B}|E_i\rangle$. This implies that $\hat{B}|E_i\rangle$ is an eigenstate of $\hat{A}$ as well with the same eigenvalue λ. In the case of non-degenerate eigenvalues, it is quite clear that this implies that $|E_i\rangle$ is also an eigenstate of $\hat{A}$. While it is a less direct proof in the case of degenerate eigenvalues, the same property still holds. We won't be proving it in this paper, however, there are many available proofs if the reader is curious. Thus, we have that $\beta \subseteq$ (Eigenbasis for $\hat{B}$), and the inverse can be shown without loss of generality for the (Eigenbasis for $\hat{B}$) $\subseteq \beta$. $\therefore$ $\hat{A}$ and $\hat{B}$ share a basis of eigenstates. This is known as a common or **simultaneous eigenbasis**.

This is an important implication since when we make an observation/reading our quantum state will collapse into a definite state and we can measure the corresponding eigenvalue. If we have two commuting observables, we can measure both physical values simultaneously. Contrarily, if $[\hat{A}, \hat{B}] = 0$, this means they cannot have a simultaneous eigenbasis.

For the sake of contradiction, suppose $[\hat{A}, \hat{B}] \neq 0$ and they do have a simultaneous basis indicated by $\beta = \{E_i\}$. We have that:

$$\hat{A} = \sum_{i=1}^{n} A_i |E_i\rangle \langle E_i|, \hat{B} = \sum_{j=1}^{n} B_j |E_j\rangle \langle E_j| \implies$$

$$\hat{A},\hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} = \sum_{i=1}^{n}\sum_{j=1}^{n} A_i B_j \left|E_i\right\rangle \left\langle E_i\right| \left|E_j\right\rangle \left\langle E_j\right| - \sum_{i=1}^{n}\sum_{j=1}^{n} A_i B_j \left|E_j\right\rangle \left\langle E_j\right| \left|E_i\right\rangle \left\langle E_i\right| =$$

$$\sum_{i=1}^{n} A_i B_i \left|E_i\right\rangle \delta_{ij} \left\langle E_i\right| - \sum_{i=1}^{n} A_i B_i \left|E_i\right\rangle \delta_{ij} \left\langle E_i\right| = 0$$

Thus, $[\hat{A},\hat{B}] = 0$, which poses a contradiction to our initial assumption that the operators don't commute, so they can't have a simultaneous eigenbasis. This means that there must always exist eigenstates of one observable that is always in a superposition of the other's eigenstates. In other words, if we have two operators $\hat{A},\hat{B}$, such that $[\hat{A},\hat{B}] \neq 0$, we are uncertain about A when measuring B and vice versa.

3.3.1 The Heisenberg Uncertainty Principle

This discussion of commutators is at the heart of the derivation of the **Heisenberg Uncertainty Principle**, which is a fundamental property of Quantum Mechanics. It is derived from the **Canonical Commutator** $[\hat{x},\hat{p}] = i\hbar$ which implies that position and momentum share no common eigenstates. The more generalized formula we commonly expect to see is $\sigma_{\hat{x}}\sigma_{\hat{p}} \geq \frac{\hbar}{2}$. This form explicitly shows that the more certain or precise our measurement of position is, the less precise or certain our measurement of momentum is. The Heisenberg uncertainty principle is a specific form of the **Generalized Uncertainty Principle** of the form $\sigma_{\hat{A}}\sigma_{\hat{B}} \geq |\frac{1}{2i}\langle[\hat{A},\hat{B}]\rangle|^2$. There are stronger uncertainty relations like the **Maccone-Pati Uncertainty Principle**, but these examples were added to show the significance of using commutators to compare observables.

4 Unitary Operators

Unitary Operators are linear operators that satisfy $\hat{U}^\dagger = \hat{U}^{-1}$. Unitary Operators preserve the Inner Product. Consider the inner product of $\langle\hat{U}\psi|\hat{U}\phi\rangle = \langle\hat{U}\hat{U}^\dagger\psi|\phi\rangle = \langle\hat{U}\hat{U}^{-1}\psi|\phi\rangle = \langle\psi|\phi\rangle$. Thus, we have the key relationship:

$$\langle\hat{U}\psi|\hat{U}\phi\rangle = \langle\psi|\phi\rangle$$

These Unitary Operators are invaluable to represent transformations on our vector space that preserve probability relations, length of our kets, and the "angle" between them, such as Rotations, Translations, and the Time Evolution Operators. Note that if we have $\hat{U}\left|\psi\right\rangle = \lambda\left|\psi\right\rangle$, where $\left|\psi\right\rangle$ is a normalized eigenvector, we have $\langle\hat{U}(\psi)|\hat{U}(\psi)\rangle = \lambda\lambda^* = |\lambda|^2 = 1$, since $\langle\psi|\psi\rangle$. This means that the eigenstates preserve their length over $\hat{U}$.

5 The Lagrangian and Generators

The quantum state is an object in Quantum Mechanics holding all the data of our wave/particle such as its possible position, momentum, energy, and probabilities. In Newtonian Mechanics, however, we work with all these dynamicals individually; because of this, these properties appear distinct. The **Lagrangian** attempts to remedy this by creating an object that has all the desired info of the Newtonian State: $\mathcal{L}(t,x(t),\dot{x}(t)) = \frac{1}{2}m\dot{x}^2 - V(x)$, where V(x) is the potential function. This form appears both in classical and quantum mechanics, where it's derived using the principle of least action and while deriving the quantum path integral respectively.

The position as a function of time can be determined by finding the function that forms a stationary point on the function:

$$S[\mathcal{L}] = \int_{t_1}^{t_2} \mathcal{L}(t,x(t),\dot{x}(t))dx$$

This is known as the **Action Integral**. To find its stationary point, we choose some $x(t)$, then plot $S[\mathcal{L}]$ as a function of how much we vary our $x(t)$. We obtain a stationary point (maxima, minima, or saddle point) when if we vary $x(t)$ slightly, our $S[\mathcal{L}]$ will barely change. This stationary point for our action integral corresponds with the path of our particle. This is what is known as the path of least/most/saddle action or much more generally the **Path of Stationary Action**.

To find such $x(t)$, we can implement tools from the calculus of variation. With this, we recognize such $x(t)$ must follow $\frac{\partial \mathcal{L}}{\partial x} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}}$ or the **Euler-Lagrange equation**. Substituting for our Lagrangian, we get:

$$\frac{\partial \mathcal{L}}{\partial x} = -\frac{d}{dx}(V(x)) = \frac{d}{dt}(m\dot{x}) = \frac{d}{dt}p = m\ddot{x} = F = ma$$

What we have just derived from the Lagrangian is the fact that the path of our particle will satisfy Newton's Second Law. The Euler-Lagrange equation is a powerful tool that states that if we vary a single parameter and fix all other parameters, we can arrive at a relationship with a physical quantity that was considered to be unrelated (in this case a change in momentum "generates" spacial change). We can do this for each of our parameters (replacing velocity with momentum and considering the angular case). Thus, we arrive at the set of equations:

$$\boldsymbol{\frac{\partial \mathcal{L}}{\partial x} = \frac{d}{dt}p} \qquad \boldsymbol{\frac{\partial \mathcal{L}}{\partial t} = -\frac{d}{dt}E} \qquad \boldsymbol{\frac{\partial \mathcal{L}}{\partial p} = \frac{d}{dt}x} \qquad \boldsymbol{\frac{\partial \mathcal{L}}{\partial \theta} = \frac{d}{dt}L}$$

These are what are known as **Generating Functions** in physics. While they are all useful in their own rights, to derive the Schrödinger Equation, we will be using the second equation: changes in energy "generate" a change in time. As mentioned before, these relationships are largely hidden but are necessary to adapt classical mechanics within our quantum framework.

6 The Schrödinger Equation

These generators hold in classical mechanics, but we must now relate them to quantum mechanics. Remember that since both the Lagrangian and the quantum state ket are singular objects that carry information about our particle, we can consider how the state ket holds in the place of the Lagrangian. As our particle advances in time, it may interact with certain fields or other particles. Hence, we expect its quantum state to change in time. We can represent this change in state with an operator $\hat{U}(t)$ called the **Time-Evolution Operator**: $\hat{U}(t)\,|\psi\rangle = |\psi(t)\rangle$.

Using intuition and properties of Quantum Mechanics, we can expect a few properties from our $\hat{U}(t)$ operator:

- $\boldsymbol{\hat{U}(0) = \hat{I}}$: When we don't have a change time, we should expect no change in our quantum state.
- **$\hat{U}(t)$ Should Be Invertible**: When we evolve our time ket, we should be able to proceed backward through physical time to obtain our original state ket, so $\boldsymbol{\hat{U}(t)}$ should have an inverse operator.
- **Total Probability Should Be Conserved**: Our operator must be norm-preserving such that probability is conserved.

As we have established previously, the Time-Evolution Operator is unitary, and these conditions we have provided, support this conclusion. Thus, we have that $\hat{U}^{-1} = \hat{U}^{\dagger}$. Finding this operator, however, is complex. What we often resort to in physics is to consider the action of our operator over an infinitesimally small time, $\lim_{\Delta t \to 0} \hat{U}(\Delta t)$. For sufficiently small t, we can consider the Taylor Expansion of $\hat{U}$ around t (with an extra term to account for the non-linear expansion; this however is unimportant since it will soon drop):

$$\lim_{\Delta t\to 0} \hat{U}(t+\Delta t) = \lim_{\Delta t\to 0}(\hat{U}(t) + \dot{\hat{U}}(t)\Delta t + \hat{O}(t)(\Delta t)^2) = \lim_{\Delta t\to 0}(\hat{U}(t) + \dot{\hat{U}}(t)\Delta t + \hat{O}(t)(\Delta t)^2) \implies$$

$$\lim_{\Delta t\to 0} \hat{U}(t+\Delta t)\,|\psi\rangle = \lim_{\Delta t\to 0}(|\psi(t)\rangle + \dot{\hat{U}}(t)\Delta t\,|\psi\rangle + \hat{O}(t)(\Delta t)^2\,|\psi\rangle) \implies$$

$$\lim_{\Delta t\to 0} \frac{|\psi(t+\Delta t)\rangle - |\psi(t)\rangle}{\Delta t} = \lim_{\Delta t\to 0}(\dot{\hat{U}}(t)\,|\psi\rangle + \frac{\hat{O}(t)(\Delta t)^2\,|\psi\rangle}{\Delta t}) = \lim_{\Delta t\to 0}\dot{\hat{U}}(t)\,|\psi\rangle \implies$$

$$\boldsymbol{\frac{d}{dt}\,|\psi(t)\rangle = \dot{\hat{U}}(t)\,|\psi\rangle}$$

Moreover, we can further analyze the unitary property:

$$\hat{I} = \hat{U}(t)\hat{U}^{\dagger}(t) \implies 0 = \dot{\hat{U}}(t)\hat{U}^{\dagger}(t) + \hat{U}(t)\dot{\hat{U}}^{\dagger}(t) \implies \dot{\hat{U}}(t)\hat{U}^{\dagger}(t) = -\hat{U}(t)\dot{\hat{U}}^{\dagger}(t) \implies$$

$$\boldsymbol{\dot{\hat{U}}(t)\hat{U}^{\dagger}(t) = -(\dot{\hat{U}}(t)\hat{U}^{\dagger}(t))^{\dagger}}$$

Therefore, we have that the product of the derivative of our Time-Evolution Operator and the Hermitian Adjoint of the Time-Evolution Operator is anti-Hermitian. This means there exists some Hermitian Operator $\hat{H} = i\dot{\hat{U}}(t)\hat{U}^{\dagger}(t) \implies \frac{1}{i}\hat{H}\hat{U}(t)\left|\psi\right\rangle = \dot{\hat{U}}(t) = \frac{1}{i}\hat{H}\left|\psi(t)\right\rangle$. Coupled with our previous expansion of $\hat{U}$, this Hermitian Operators implies that we have a combined equation of the form

$$i\frac{d}{dt}\left|\psi(t)\right\rangle = \hat{H}\left|\psi(t)\right\rangle$$

However, this still leaves ambiguities as to the definition of $\hat{H}$, but we can use our intuition gained from generating functions to understand that this must be the Hamiltonian Operator for energy since energy generates temporal change and the Hamiltonian satisfies the condition of being Hermitian.

Now, we must consider the units of our equation. $i\frac{d}{dt}\left|\psi(t)\right\rangle$ has units of s^{-1}, and the Hamiltonian has units of energy, J. Hence, we are looking for some constant c with units $J \cdot s$. This constant, however, is more than just used to reconcile units. From our understanding of Quantum Mechanics, we know that energy is found in quanta that is proportional to a wave's angular frequency by its reduced Planck's constant: $E = \hbar\omega$ with constants $J \cdot s$. Therefore, we predict that this is our unknown constant. This was later verified in experiments. While this derivation utilizes much more of a heuristic understanding of physics, it shows the robustness of our mathematical framework. Hence, we have our final equation, the Time-Dependent Schrödinger Equation:

$$i\hbar\frac{d}{dt}\left|\psi(t)\right\rangle = \hat{H}\left|\psi(t)\right\rangle$$

7 Conclusion

In this paper, we have explored the mathematical foundations underpinning Quantum Mechanics, with a particular focus on the pivotal role of Linear Algebra in shaping its framework. By examining the properties and structures of Hilbert Spaces, Linear Operators, and their intricate relationships, we have outlined the essential constructions necessary for understanding quantum systems. While a bit lengthy, this project underscored the foundational interplay between these mathematical tools and the principles of the quantum world. Thank you for reading.

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